Arista Access Node Optical Transport Failure SOP (ALARM_5710)

Node: YEL-ACC-32 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_5710

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve a critical optical transport failure in an Arista access node, specifically the YEL-ACC-32 node at the Yelahanka Sub-station. The failure is attributed to the degradation of SFP/XFP modules or fiber cuts, resulting in a loss of receive (Rx) and transmit (Tx) power. The Arista access node, operating in a high availability environment, utilizes the eAPI protocol for monitoring and management. The control plane, responsible for routing and forwarding decisions, relies on the data plane to implement these decisions. The data plane, consisting of the ASIC, buffers, and fiber optic interfaces, handles the actual packet forwarding. In this failure scenario, the data plane's ability to forward packets is compromised due to the optical transport failure, impacting the overall network performance and availability. The root cause of this failure could be attributed to various factors, including SFP/XFP degradation, fiber cuts, or optical signal degradation. This SOP outlines the steps to diagnose and resolve these root causes, ensuring minimal downtime and optimal network performance.

Pre-requisite Checks

- Verify the site location and node ID to ensure the correct access node is being addressed.
- Check the redundant link status to confirm the presence of a backup link.
- Dispatch remote hands to the Yelahanka Sub-station if necessary.
- Ensure the Arista device is accessible via the eAPI protocol.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Optical signal degradation	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces status, Expected Output: Interface Status Description, Description: This command displays the status of all interfaces, including the SFP/XFP modules, and helps identify if any of the interfaces are down or degraded.

Command: show interfaces detail, Expected Output: Interface Status Description, Description: This command provides detailed information about each interface, including the receive and transmit power levels,
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and helps identify if there are any issues with the SFP/XFP modules or fiber optic interfaces.

Command: show optics, Expected Output: Optics Status Description, Description: This command displays the status of the optics, including the SFP/XFP modules, and helps identify if any of them are degraded or faulty.
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Resolution Steps

Step 1: Step: Replace the degraded SFP/XFP module, Command: install SFP-XFP module , Description: Replace the degraded SFP/XFP module with a new one, ensuring the correct type and compatibility with the Arista access node.

Step 2: Step: Check and repair the fiber optic cables, Command: show interfaces status, Description: Verify the status of the fiber optic cables and repair any damaged or cut cables to ensure optimal network performance.

Step 3: Step: Adjust the optical signal settings, Command: Configure optics settings, Description: Adjust the optical signal settings to optimize the transmission and reception of packets, ensuring minimal signal degradation and optimal network performance.

Post-Check Verification

```
Command: show interfaces status, Expected Output: Interface Status Description, Description: Verify the status of all interfaces, including the SFP/XFP modules, to ensure they are up and functioning correctly.

Command: show interfaces detail, Expected Output: Interface Status Description, Description: Verify the status of each interface, including the receive and transmit power levels, to ensure they are within the normal operating range.

Command: show optics, Expected Output: Optics Status Description, Description: Verify the status of the optics, including the SFP/XFP modules, to ensure they are functioning correctly and without any degradation.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: remove SFP-XFP module , Description: Restore the original SFP/XFP module that was replaced during the resolution process, ensuring the correct type and compatibility with the Arista access node.

- Step: Restore the original fiber optic cables, Command: Verify fiber optic cables, Description: Verify the status of the fiber optic cables and restore the original cables if necessary, ensuring optimal network performance.

- Step: Restore the original optical signal settings, Command: Restore optics settings, Description: Restore the original optical signal settings to ensure minimal signal degradation and optimal network performance.